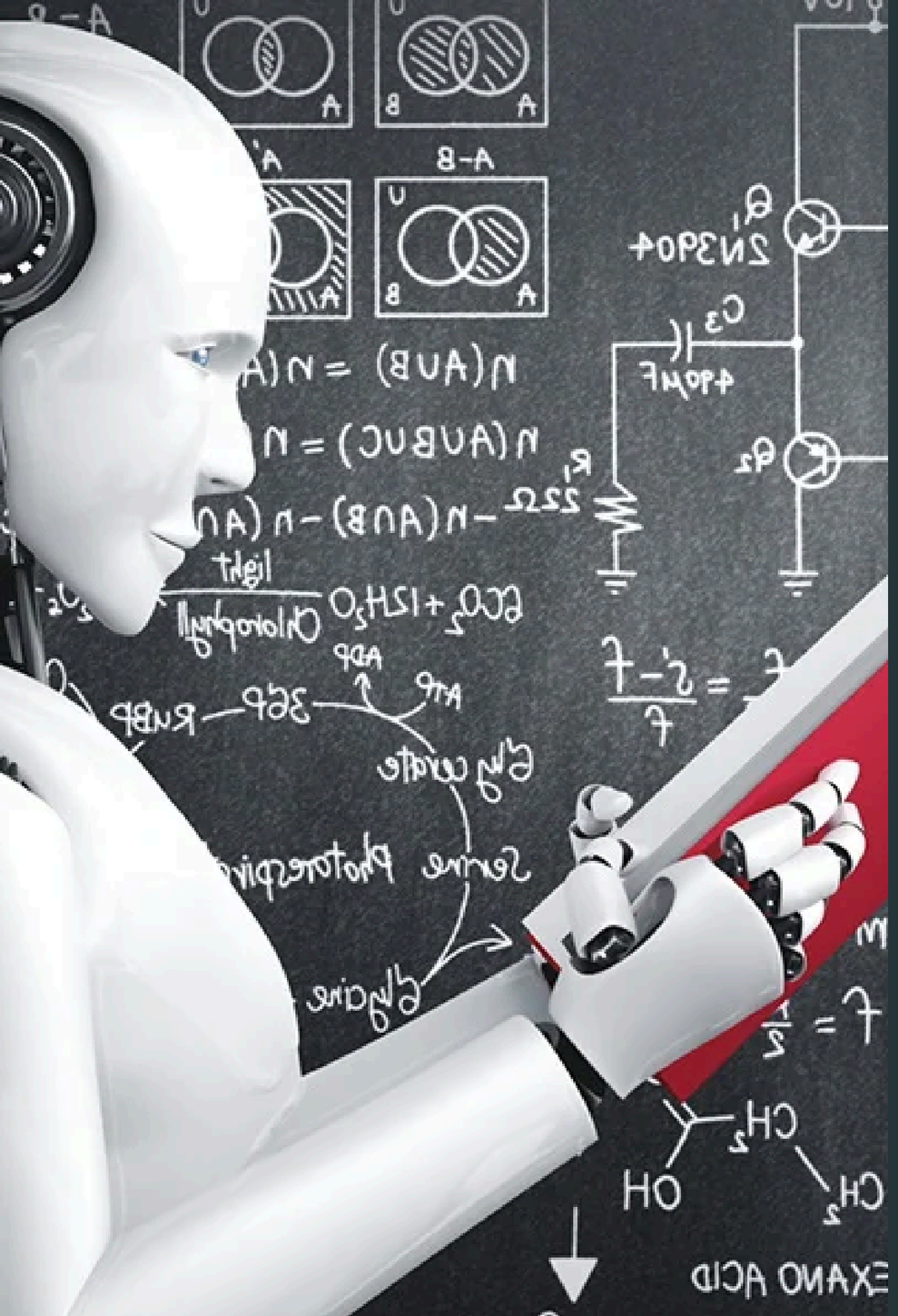

BRACT's Vishwakarma Institute of Information Technology, Pune

Department of Computer Science and Engineering (Artificial Intelligence)

Project Title - DSA Forge : AI Teaching Assistant

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Project Guide - Dr. Nilesh Sable Sir

Problem Statement

Design and Development of a AI Teaching Assistant Using Generative AI

Many college students struggle with complex subjects due to differences in prior knowledge, learning pace, and lack of personalized guidance. Traditional classroom teaching and static learning resources often fail to adapt to individual student needs, making it difficult for students to identify their weak areas and improve effectively.

The proposed solution aims to support college students by providing personalized, interactive, and adaptive learning assistance using Generative AI, thereby improving comprehension, confidence, and academic performance in technically challenging subjects like DSA.

LITERATURE SURVEY

Paper Title	Author	Work	Limitation
Local Generative AI Teaching Assistant (CE-GAITA)	Jing-Wen Wu et al.	Proposes a locally deployed Generative AI teaching assistant using a Retrieval-Augmented Generation (RAG) architecture that allows instructors to upload course materials and provides accurate, privacy-preserving question-answering support for campus learning environments.	The system mainly focuses on content retrieval and Q&A accuracy and does not incorporate student-level personalization, adaptive question generation, or continuous learning progress tracking.
TutorLLM: Customizing Learning Recommendations with Knowledge Tracing and Retrieval-Augmented Generation	Li, Zhaoxing, et al.	TutorLLM proposes a personalized learning system that integrates knowledge tracing with Retrieval-Augmented Generation (RAG) to tailor AI responses according to a student's learning state. The model tracks learner progress over time and uses curriculum-relevant retrieval to generate context-aware explanations and recommendations, aiming to improve individualized support and adaptive feedback during learning.	While TutorLLM advances personalization and context-grounded responses, it primarily focuses on learner modeling and content retrieval without a complete implementation of **adaptive question generation**, **real-time difficulty adjustment based on student performance**, or **comprehensive progress tracking dashboards** that guide ongoing practice and mastery — particularly for technically rigorous areas like Data Structures and Algorithms
Generative AI for Education: A Retrieval-Augmented System for Effective Feedback in Self-Assessment	Martinez-Romo, Juan, et al.	The paper presents a **Retrieval-Augmented Generation (RAG)–based educational system** that provides accurate, curriculum-grounded feedback during student self-assessment. By retrieving content directly from course textbooks and using prompt engineering, the system generates personalized explanations that correct misconceptions and reinforce key concepts, improving the quality and consistency of AI-generated feedback.	The system primarily focuses on **feedback generation during self-assessment** and does not incorporate **adaptive question generation, continuous learner profiling, real-time difficulty adjustment, or long-term progress tracking**, limiting its use as a full-fledged personalized teaching assistant for subjects like Data Structures and Algorithms.
AI-driven formative assessment and adaptive learning in data-science education: Evaluating an LLM-powered virtual teaching assistant	Anaroua, Fadjimata I., et al.	The paper introduces **VITA**, an LLM-powered virtual teaching assistant designed for **conversational tutoring and formative assessment** in data-science education. The system evaluates student responses, supports adaptive learning pathways, and compares the effectiveness of **RAG-based architectures with traditional LLM tutoring**, highlighting improvements in feedback quality and learning analytics.	The system is domain-specific to data science and does not focus on syllabus-aligned knowledge bases, adaptive question generation, or detailed learner profiling for technically intensive subjects like Data Structures and Algorithms
Adapting Gamified Learning Systems Using Educational Data Mining Techniques	Lamya F Daghestani, Lamiaa F Ibrahim, Reem S Al-Towirgi, Hesham A Salman	The system relies on **behavioral data mining and rule-based adaptation** and does not incorporate **Generative AI, conversational tutoring, real-time conceptual explanations, or adaptive question generation**, limiting personalized instructional support for complex problem-solving tasks in DSA.	Lacks GenAI-driven explanations, conversational tutoring, and adaptive problem generation.

Proposed Methodology

1 User Entry & Central Dashboard

- Users start from a single entry point
- Dashboard acts as the control hub
- Connects all modules and manages navigation + feedback loops

2 Learning Path Selection

- Topic selection (Arrays, DP, etc.)
- Multiple difficulty levels (Beginner → Advanced)
- AI-generated quizzes to progress between levels

3 AI-Assisted Learning

- AI chat assistant for guidance
- Automated code reviews with feedback
- Dynamic problem generation
- AI-based evaluation of solutions

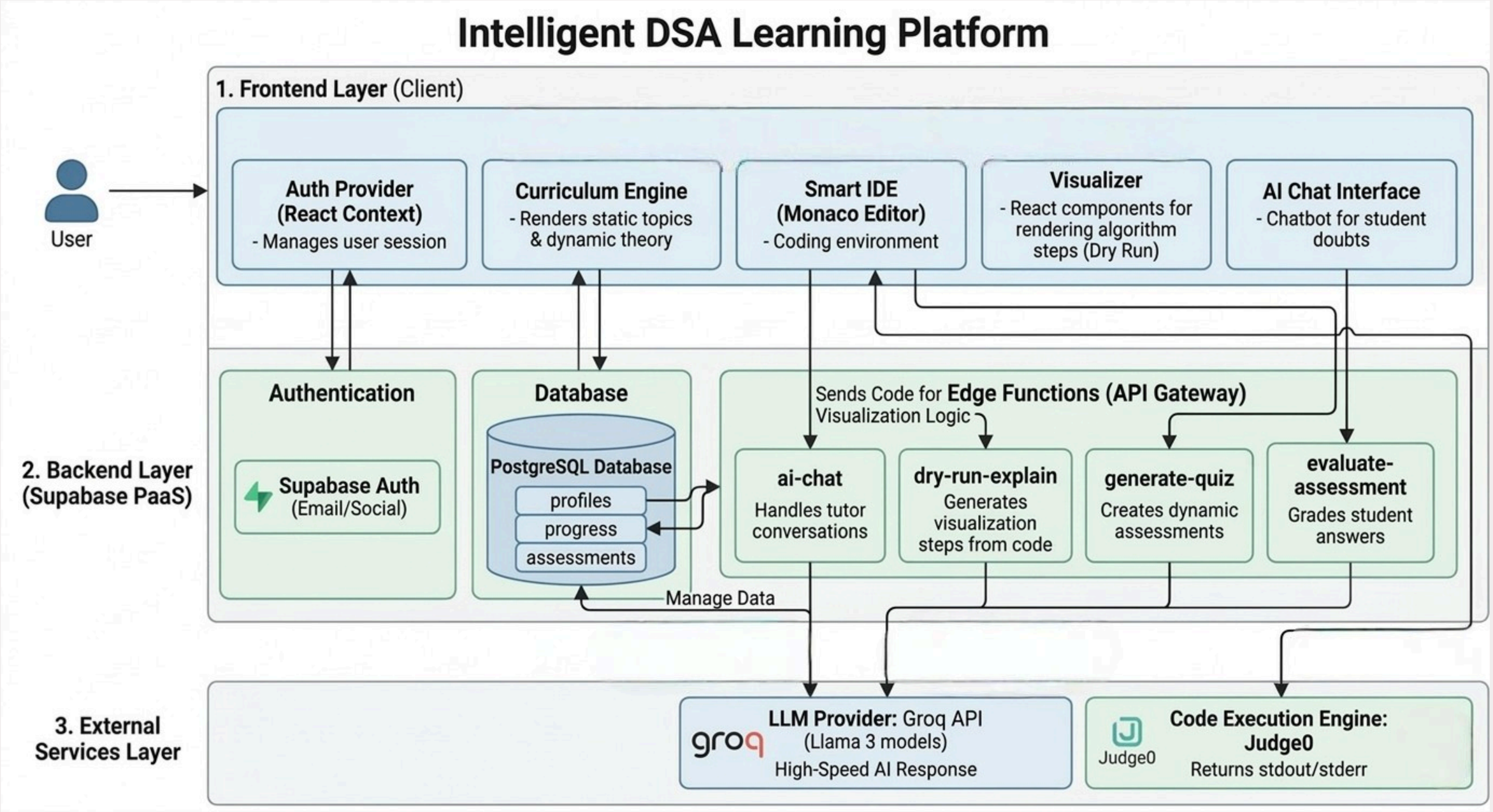
4 Practice & Coding Environment

- Built-in code editor
- Run/submit code with test cases
- Hints and dry-run visualizations
- Instant feedback on submissions

5 Progress Tracking & Personalization

- Performance analytics (charts, streaks)
- Skill proficiency updates
- Personalized next-step recommendations
- Solution walkthroughs with explanations

System Architecture



Tech Stack

Frontend: React.js, Tailwind CSS, shadcn/ui + Radix UI

Backend: Supabase (PostgreSQL, Auth, Edge Functions)

LLM Framework : Qwen-3-32B and Code Llama for Coding Guidance

Code Editor: Monaco Editor (code editor)

Targeted Impact

01

Improving Learning Outcomes & Student Confidence

02

Preparing Students for placements

03

Enhancing Teacher and Educational Effectiveness

Thank You